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AWS EC2 SQL Instance Health Check — Key Steps for Optimal Performance

When performing a **health check for SQL Server running on an AWS EC2 instance**, I follow these steps to ensure the **database, instance, and underlying infrastructure** are functioning efficiently:

1. EC2 Instance-Level Checks

- Verify the **EC2 instance state** (running, healthy) from AWS Console or CLI.
- Check **instance type**, CPU credits (for burstable types like t3/t4), and resource utilization.
- Ensure correct **EBS volume type and size** (gp3/io1 for SQL workloads).
- Review **CloudWatch metrics** for CPU, memory, disk, and network I/O trends.
- Confirm **OS-level performance** using Task Manager (Windows) or top/iostat (Linux).
- Validate **Elastic IP or DNS configuration** for connectivity consistency.

2. SQL Server Service Verification

- Check that all essential **SQL Server services** (Engine, Agent, SSIS, SSRS if applicable) are running.
- Review **SQL Server and Windows Event Logs** for errors or warnings.

3. Storage & Disk Health

- Check **disk latency and throughput** metrics in CloudWatch.
- Ensure adequate **free space** on data, log, TempDB, and backup drives.
- Confirm **EBS optimization** is enabled for SQL storage volumes.

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4. Database Status & Consistency

- Validate that all **databases are online** and accessible.
- Run **DBCC CHECKDB** to verify logical consistency.
- Monitor **TempDB health** (file count, size, contention).

5. Security & Access Control

- Review SQL Server logins and **failed login attempts**.
- Verify **Windows Firewall and AWS Security Group rules** allow only approved IPs and ports (default 1433).
- Check **IAM roles/policies** attached to the instance are least-privileged.

6. Backup & Recovery Validation

- Confirm scheduled **SQL Agent backup jobs** are successful.
- Verify **backup files** are stored in S3 (if applicable) or attached EBS volumes.
- Test a **sample restore** from backup to ensure recoverability.

7. Performance & Query Optimization

- Review **wait statistics**, **blocking sessions**, and **long-running queries**.
- Analyze **execution plans** for optimization opportunities.
- Ensure **indexes and statistics** are maintained regularly.
- Check **MAXDOP**, **Cost Threshold for Parallelism**, and **Memory Settings**.

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🌐 8. High Availability & DR Checks

- If **Always On Availability Groups** are configured:
  - Verify **replica synchronization**, **listener status**, and **failover readiness**.
- Confirm **automated backups** and **cross-region snapshot replication** (if used).

🧾 9. Monitoring, Alerts & Maintenance

- Review **CloudWatch Alarms**, **AWS SNS notifications**, or third-party tools (SolarWinds, Redgate, etc.).
- Validate **SQL Agent job success/failure** history.
- Review scheduled **maintenance plans** for backups, index rebuilds, and integrity checks.

💸 10. Cost & Optimization Review

- Check **EC2 utilization reports** to ensure right-sizing.
- Consider **Reserved Instances** or **Savings Plans** for cost efficiency.
- Review **storage lifecycle policies** for old backups/logs in S3.

🧠 Summary

Regular EC2 SQL Instance health checks combine both **AWS infrastructure monitoring** and **SQL Server diagnostics**. This ensures high availability, strong performance, optimized cost, and full recoverability in production environments.

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